Reg. No. | | | | | | | | | | | [18EC34]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2019
NETWORK THEORY

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions. Question number 1 and 2 are compulsory and each question carries 14 marks

ii. Missing data may be suitably assumed.

1. a. Determine the equivalent resistance across the terminals A-B using star delta transformation technique for the network shown in Fig. 1 (a). **07**

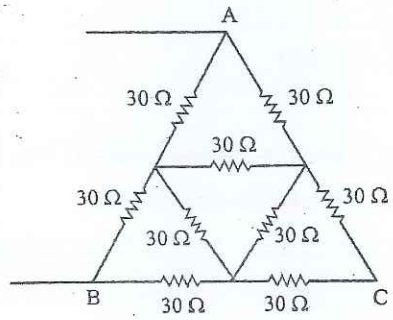

Fig. 1(a)

b. Using node voltage analysis, determine the voltage V_x in the circuit shown in Fig.Q1(b), such that the current flowing through (2+j3)Ω impedance is zero. **07**

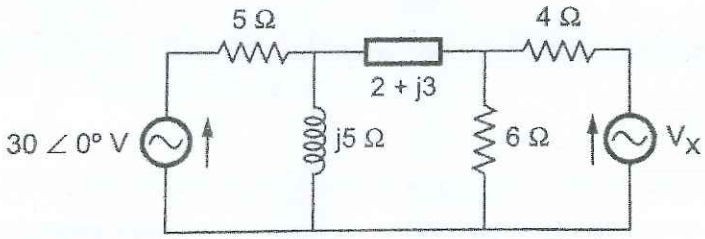

Fig. 1(b)

2. a. For the circuit shown in Fig. Q 2 (a), find the current 'i_x' and thus verify Reciprocity Theorem. **07**

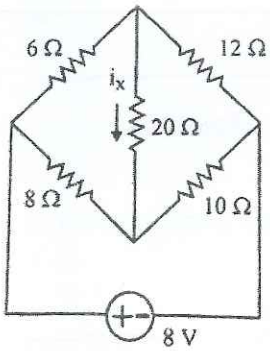

Fig. 2 (a)

b. Draw the Thevenin's and Norton's equivalent circuits across the terminals 'a' and 'b' for the circuit shown in Fig. 2(b). **07**

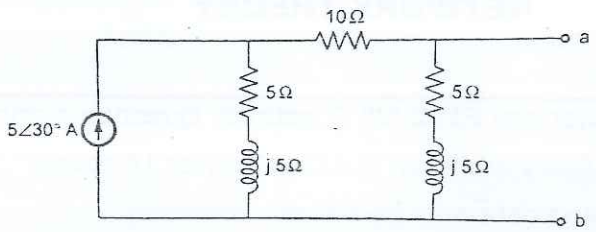

Fig. 2(b)

3 a. Find the initial and final values of the following functions: **07**

i. $i(t) = 3e^{-t} - e^{-2t}$

ii. $f(t) = 1 + e^{-t}(\sin t + \cos t)$

iii. $x(t) = u(t)$

b. Using the properties of Laplace Transformation, obtain the Laplace Transformation of the waveform shown in Fig. Q 3(b). **07**

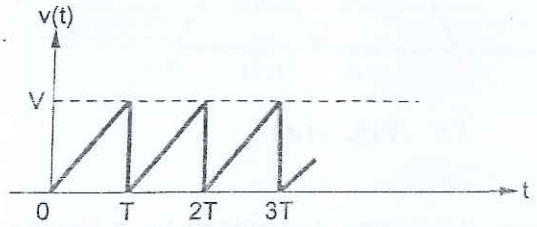

Fig. 3(b)

OR

4 a. Find the time domain signal for the following functions: **07**

i. $F(s) = \frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6}$

ii. $F(s) = \frac{4s + 5}{(s-1)^2 (s+2)}$

b. In the circuit shown in Fig. Q 4 (b), if the capacitor is initially charged to 1 V, using Laplace transformation, find an expression for i(t), when the switch K is closed at t = 0. **07**

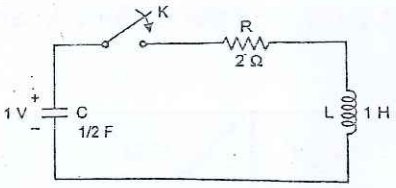

Fig. 4(b)

5 a. Obtain the values of L for which the circuit shown in Fig. Q 5(a) resonates at ω=5000rad/sec. **07**

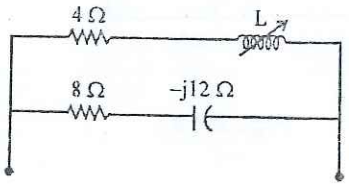

Fig. 5(a)

b. A coil of resistance 10 Ω and inductance of 0.5 H is connected in series with a capacitor. On applying a sinusoidal voltage, the current is maximum, when the frequency is 50 Hz. A second capacitor is connected in parallel with the circuit. What capacitance must it have, so that, the combination acts as a non-inductive resistor at 100 Hz? Calculate the total current supplied in each case, if the applied voltage is 220 V. **07**

OR

6 a. Derive an expression for resonant frequency of the circuit shown in fig.6(a) **07**

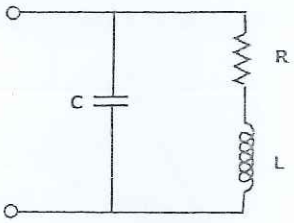

Fig.6(a)

b. An inductive coil of resistance 6 Ω and inductance 1mH is connected in parallel with another branch consisting of a resistance of 4 Ω in series with a capacitance of 20μF. Find the resonant frequency and the corresponding current, when the applied voltage is 200 V. **07**

7. a. Define admittance parameters. Represent admittance parameters in terms of hybrid parameters. **07**

b. The Z parameters of a two port network are $Z_{11}=20\Omega$, $Z_{12}=10\Omega$, $Z_{21}=10\Omega$ and $Z_{22}=30\Omega$. Find its Y and ABCD Parameters. **07**

OR

8 a. Define impedance parameters. Represent impedance parameters in terms of transmission parameters. **07**

b. The 'h' parameters of a certain two-port network are $\begin{vmatrix} 5 & 2 \\ 3 & 6 \end{vmatrix}$. Determine the corresponding **07**

i. Y parameters.

ii. Transmission Parameters.